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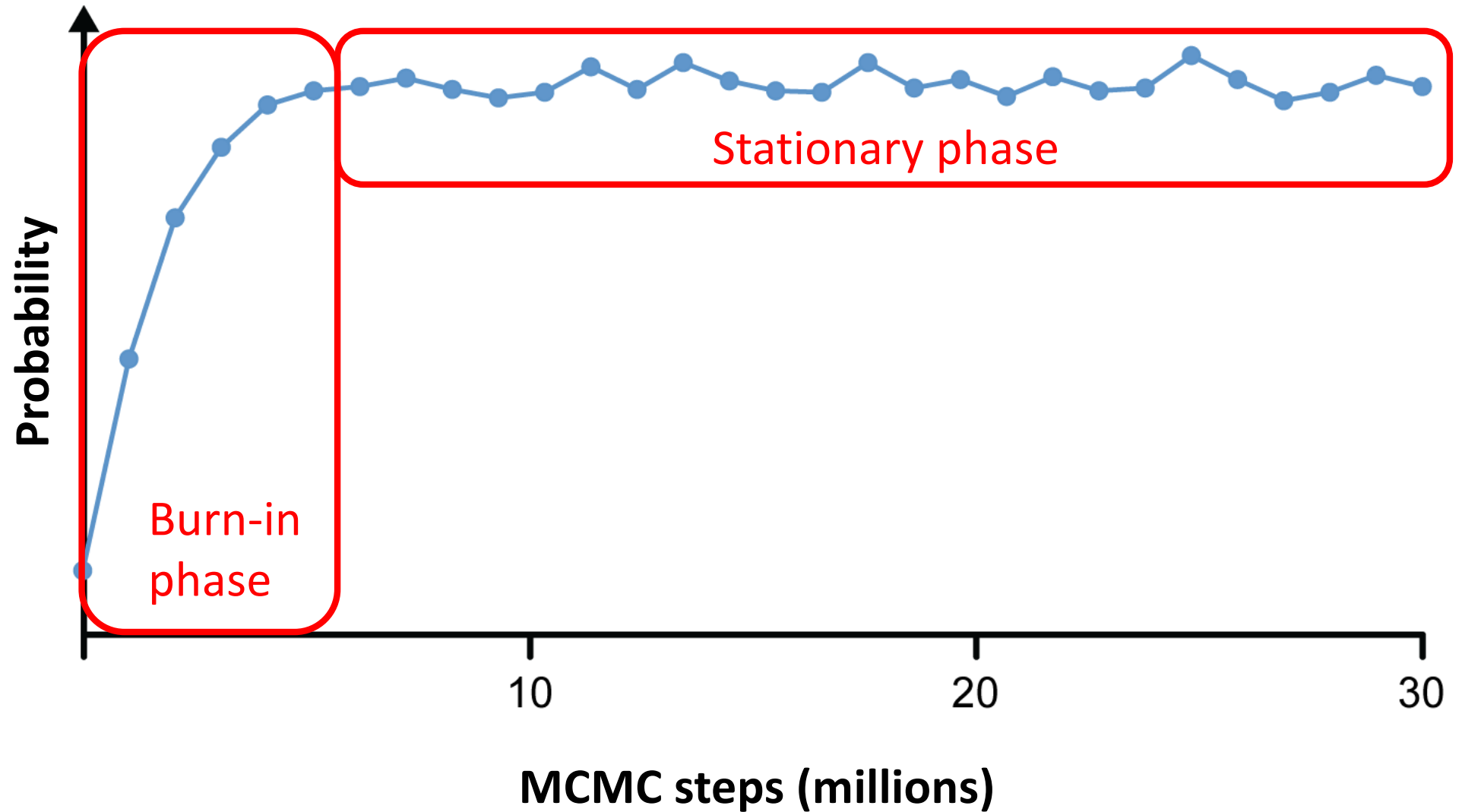
Lecture 2.3

# **Bayesian Phylogenetics II**

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# MCMC Diagnostics

# Samples from the MCMC



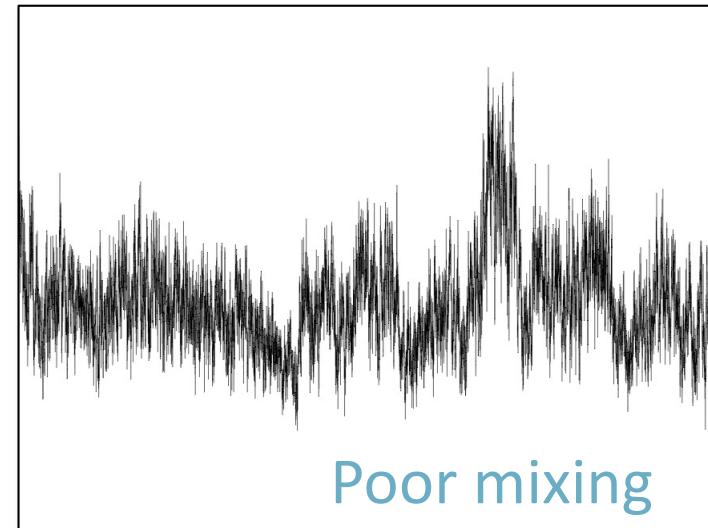
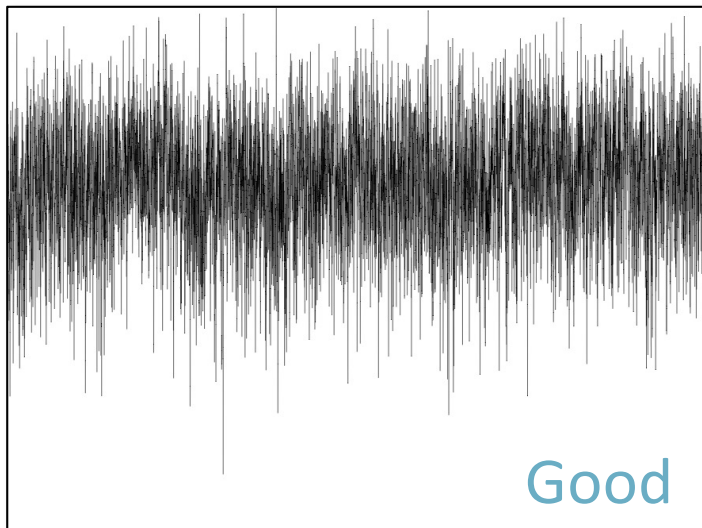
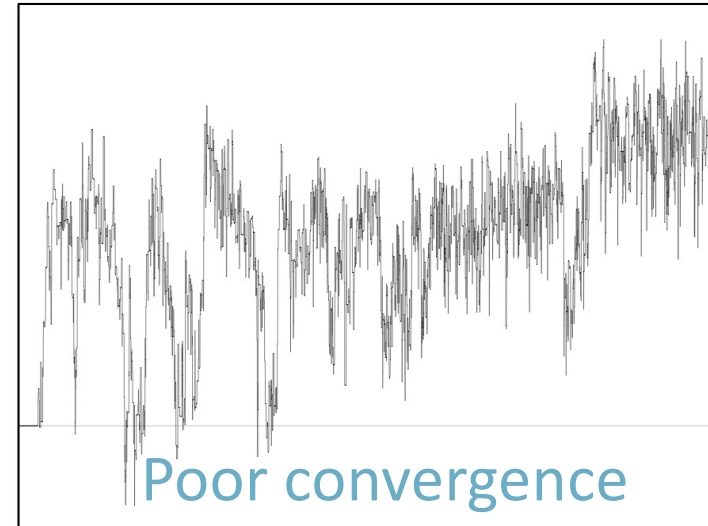
# Diagnostics

## 1. Convergence

Are we drawing samples from the stationary distribution?

## 2. Sufficient sampling

Have we drawn enough samples to allow a reliable estimate of the posterior distribution?



# Convergence

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- Run at least 2 (preferably more) independent chains
- Likelihoods should be similar
- Estimates of model parameters should be similar

# Sufficient sampling

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- **Effective sample size (ESS)**

Have we drawn enough independent samples to produce a reliable estimate of the posterior distribution?

- ESS is preferably **>200** for each parameter
- ESS can be increased by:
  - Increasing the length of the MCMC  
(and decreasing the frequency of sampling accordingly)
  - Modifying the MCMC proposals

# Advantages and Problems

# Advantages

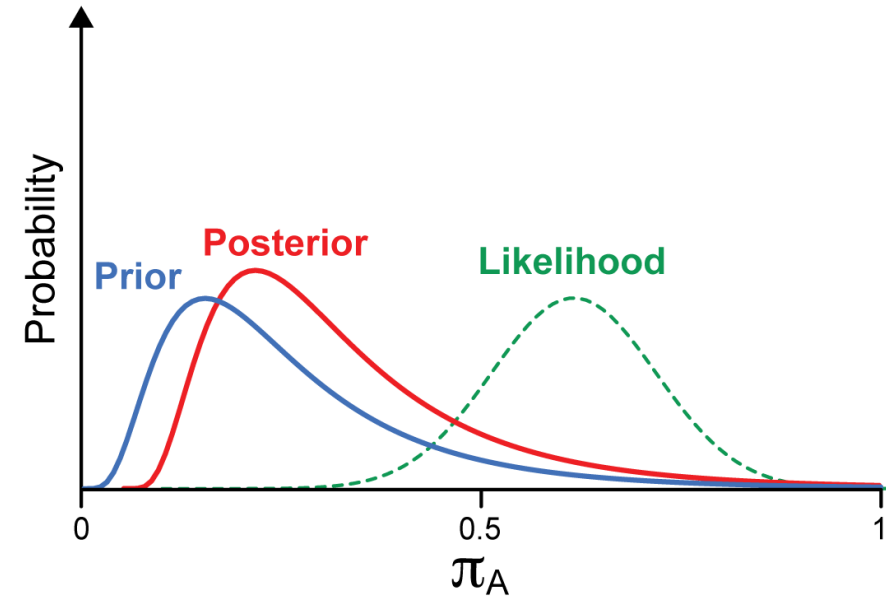
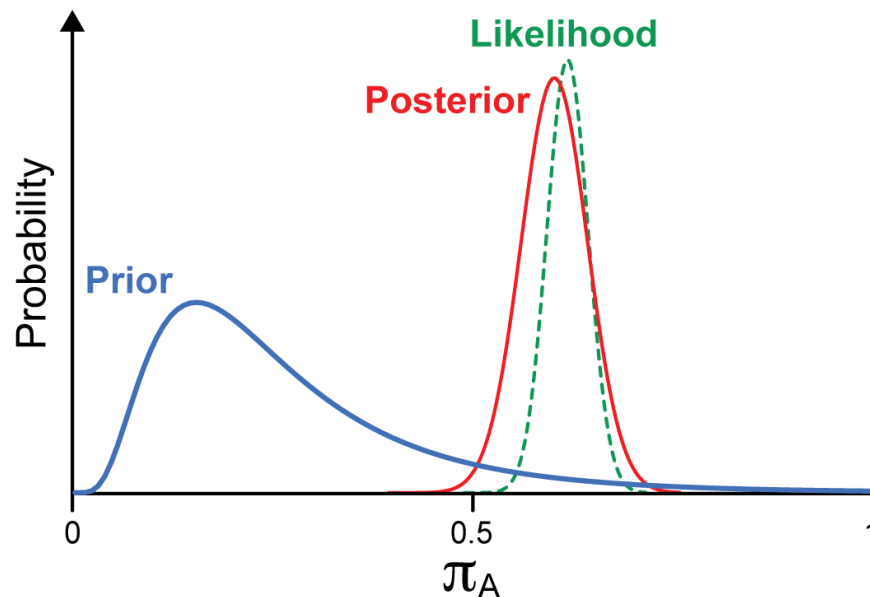
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- Able to implement complex (highly parameterised) models
- Posterior probabilities have an intuitive interpretation
- Can incorporate independent information (in the prior)
- Integrate over 'nuisance' parameters



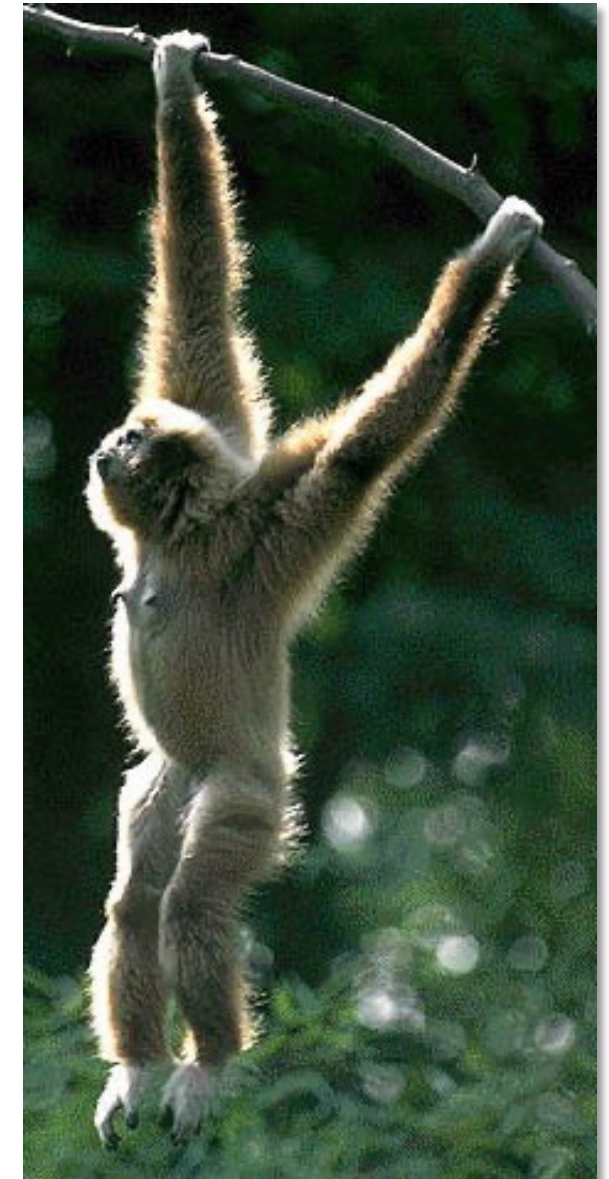
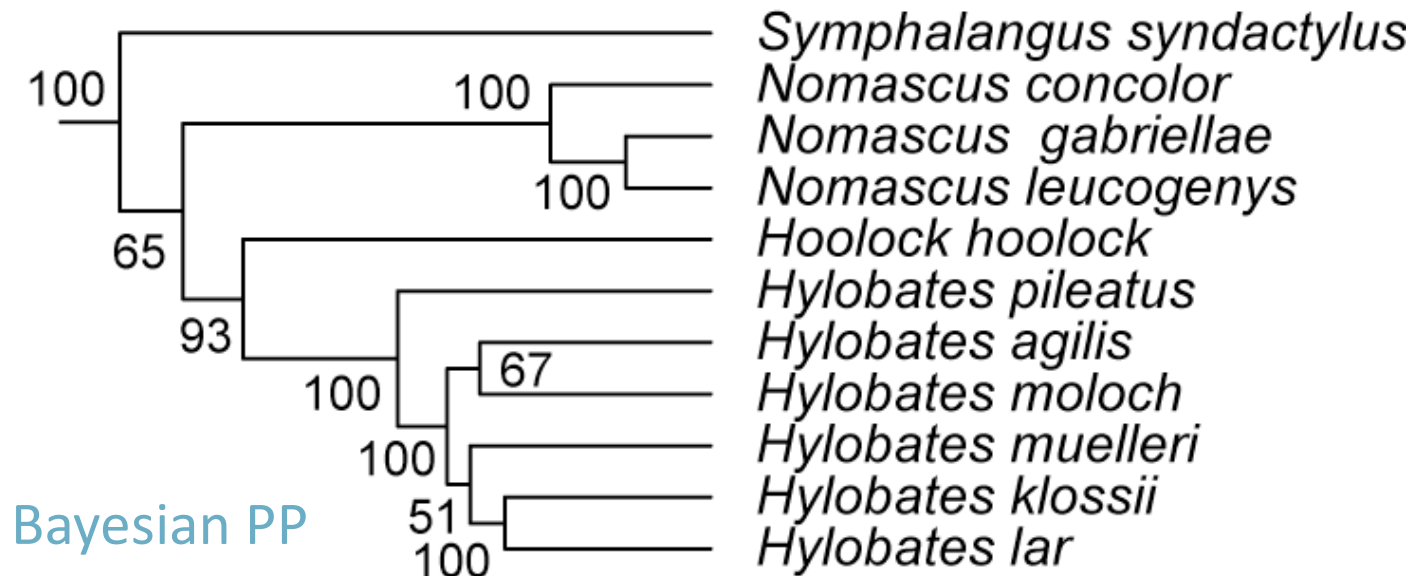
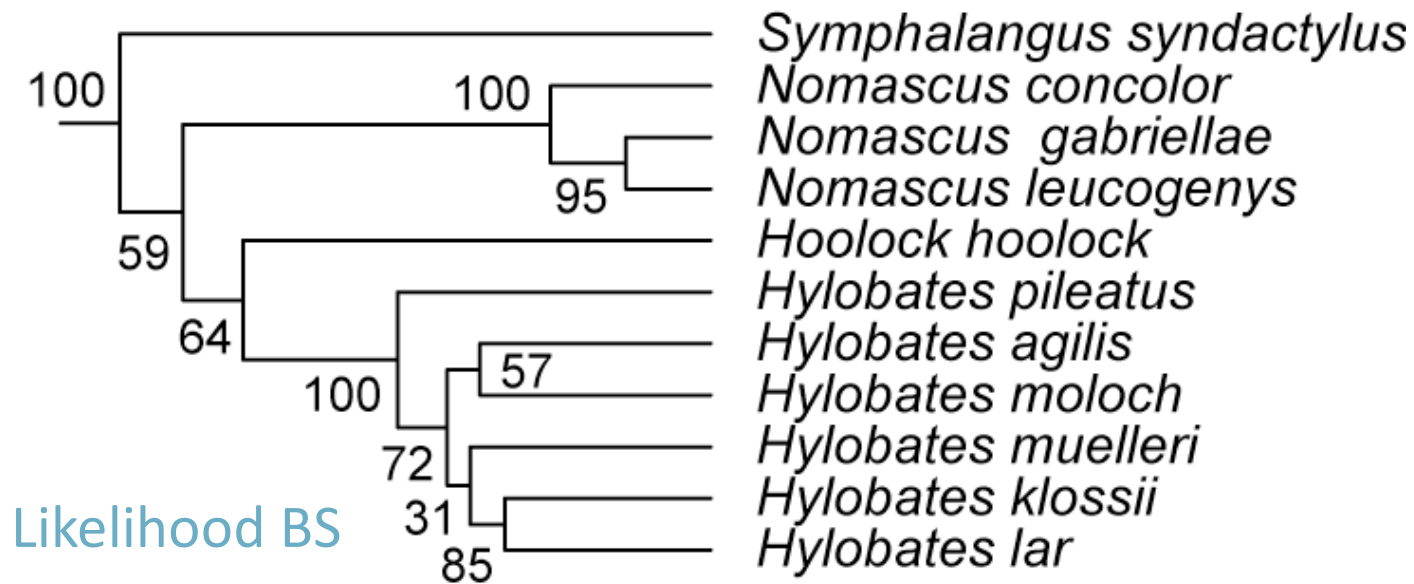
# Influence of priors

- Sensitivity of the posterior to the prior
- This problem can occur if the data are uninformative, the prior is strong, or both

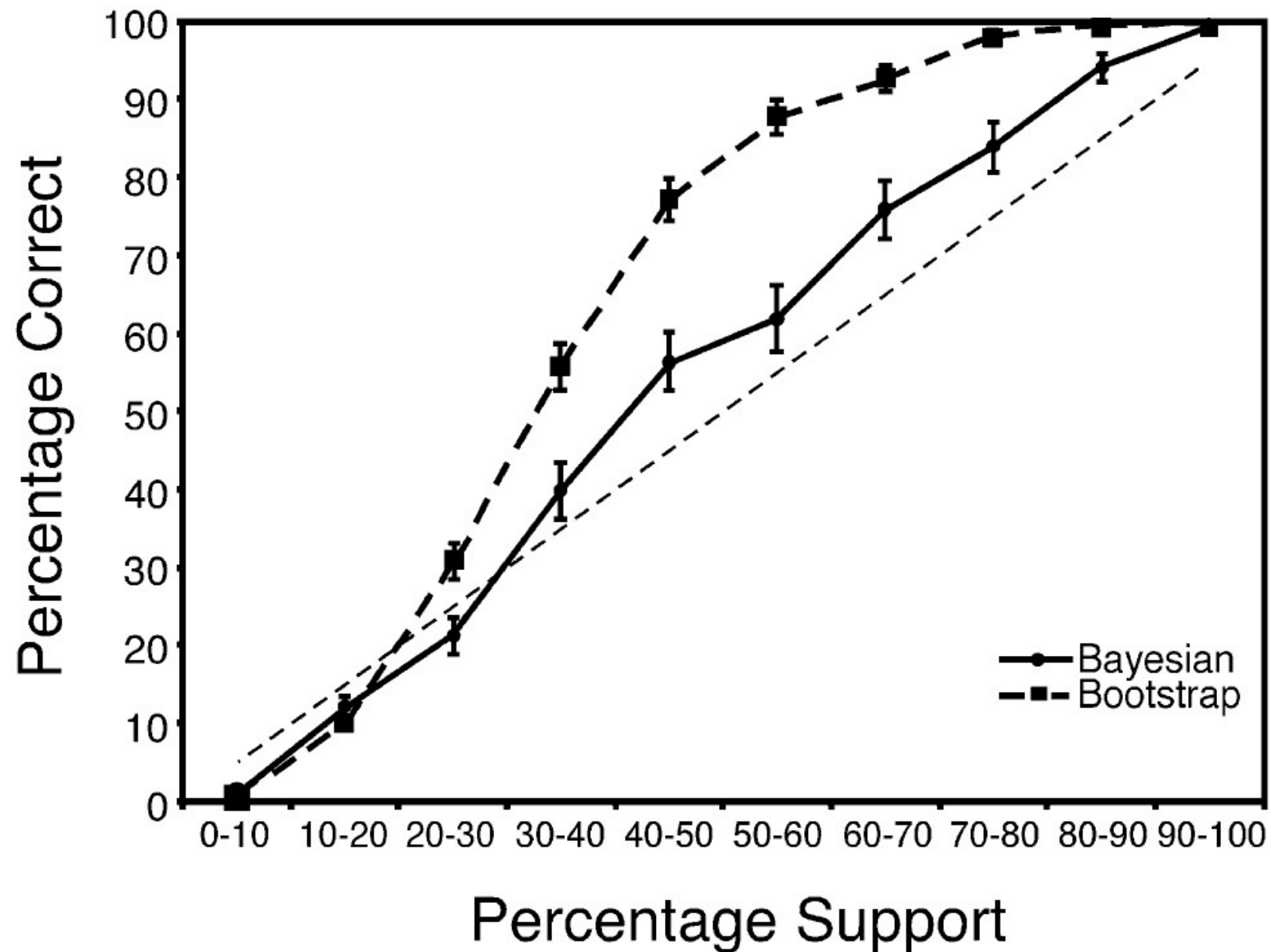


Can investigate by sampling from the prior

# Node support



# Node support



# Bayesian Model Selection

# Bayesian model selection

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- Bayesian model selection is usually based on the marginal probability of the data, conditioned on the model:

$$\Pr(\mathbf{D} | \mathbf{M})$$

- This is a weighted average of the likelihood
- Weights are given by the prior distribution

**Marginal likelihood of the model**

# Bayesian model selection

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- Pairwise comparison of models using Bayes factors
- Bayes factor is the ratio of the marginal likelihoods of the two models

$$\mathbf{BF} = \frac{\Pr(\mathbf{D} \mid \mathbf{M}_1)}{\Pr(\mathbf{D} \mid \mathbf{M}_2)}$$

$$\log \mathbf{BF} = \log \Pr(\mathbf{D} \mid \mathbf{M}_1) - \log \Pr(\mathbf{D} \mid \mathbf{M}_2)$$

- Models do not need to be nested
- Do not need to correct for number of parameters

# Bayesian model selection

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- Interpreting the Bayes factor

| <b><math>BF</math></b> | <b><math>\log BF</math></b> | <b>Evidence against <math>M_2</math></b> |
|------------------------|-----------------------------|------------------------------------------|
| 1 – 3                  | 0 – 1                       | Not worth mentioning                     |
| 3 – 20                 | 1 – 3                       | Positive                                 |
| 20 – 150               | 3 – 5                       | Strong                                   |
| > 150                  | > 5                         | Very strong                              |

# Software for Bayesian Phylogenetics





# *BEAST 1*

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- **B**ayesian **E**volutionary **A**nalysis by **S**ampling **T**rees
- Analyse population- or species-level data
- Simultaneous estimation of tree and node times
- Range of clock models
- Range of tree priors and demographic models

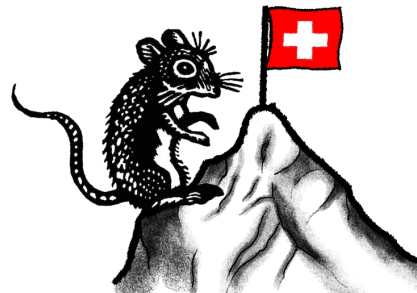


# Beast2

Bayesian evolutionary analysis by sampling trees

- Re-write of *BEAST* to increase modularity
- Users can extend *BEAST* by adding packages
- Additional tree priors not available in *BEAST* 1
- Capacity to perform simulations

For a comparison of *BEAST* 1 and 2:  
[www.beast2.org/beast-features](http://www.beast2.org/beast-features)



# *MrBayes*

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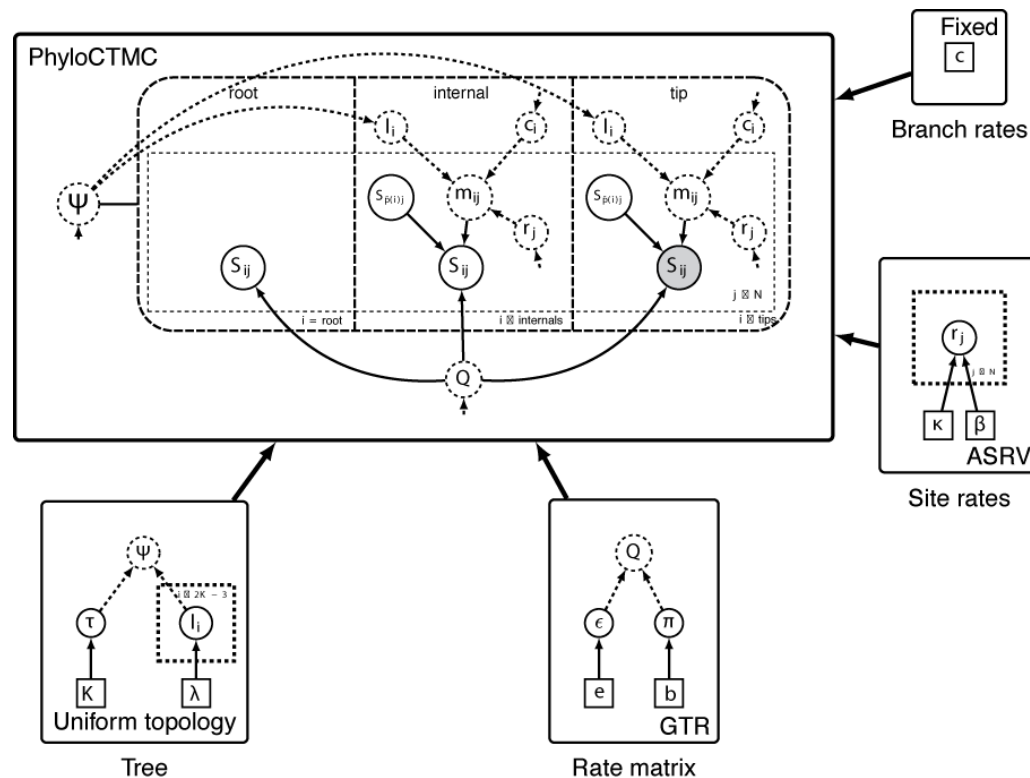
- Primarily designed for species-level data
- Simultaneous estimation of tree and node times
- Range of clock models
- Range of tree priors
- Multiple chains and MCMC diagnostics



# RevBayes



- Uses its own R-like language, Rev
- Interactive construction of graphical model
- Flexible and can be used for simulation and inference



# Useful references

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